

Miau-BOT: A Haptic Creature Design for Enhancing Emotional Regulation in Driving Scenarios

Yueyao Jiang*
Tsinghua University
Beijing, China
1822477591@qq.com

Xiwen Yao*
Tsinghua University
Beijing, China
yxw20@mails.tsinghua.edu.cn

Yuxin Zhu*
Tsinghua University
Beijing, China
zhu-yx24@mails.tsinghua.edu.cn

Xinjun Gong
Tsinghua University
Beijing, China
gong-xj24@mails.tsinghua.edu.cn

Zhihao Yao
Tsinghua University
Beijing, China
yaozh_h@outlook.com

Yut Wong
Tsinghua University
Beijing, China
wells30078899t@qq.com

Yijie Guo
Tsinghua University
Beijing, China
guoyijie.sh@gmail.com

Haipeng Mi†
Tsinghua University
Beijing, China
haipeng.mi@acm.org



Figure 1: (a) The Miau-BOT steering wheel demonstrates tactile feedback patterns for fatigue and tension emotions. (b) The design of Miau-BOT mimics the features and behavior of a cat's paw. (c) Participants interact with the tactile feedback steering wheel in a simulated driving environment.

Abstract

“Haptic creatures” are robots that simulate animal-like behaviors through touch to convey emotions, demonstrating significant potential in regulating negative emotions. Currently, methods for addressing drivers’ negative emotions primarily rely on visual, auditory, and olfactory feedback, with limited exploration of tactile feedback for emotional regulation. This study introduces Miau-BOT, a pneumatic haptic creature robot for driving scenarios. Miau-BOT provides tactile feedback through soothing dots and alerting spikes based on the driver’s state. A user study compared Miau-BOT with dot-matrix tactile feedback, exploring the unique effects

of pet-inspired designs on negative emotion regulation and emotional companionship. Results show Miau-BOT provides stronger companionship. Interviews revealed that participants developed a personal tactile identity for Miau-BOT and experienced emotional transference through metaphor. Notably, the emotional regulation effects were more pronounced in clearer tension-inducing event scenarios. This exploration highlights the role of haptic creatures in regulating negative emotions in driving contexts, offering insights for future design applications.

CCS Concepts

• **Human-centered computing** → **Haptic devices**; *Empirical studies in HCI*.

Keywords

Haptic creature; Haptic interfaces; Affective Haptic; Tactile sensors; Steering wheel; Attachment; Emotional transfer; Metaphor;

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*These authors contributed equally to this research.

†Corresponding author.

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1 INTRODUCTION

With the advancement of automotive technology, the driving experience has expanded beyond basic safety requirements to include emotional support. Studies have shown that the enclosed driving environment often leads to negative emotions in drivers [29, 48]. These emotions manifest in different scenarios, such as fatigue and distraction caused by monotonous road conditions, and tension induced by complex traffic situations [48, 51]. Such negative emotions significantly impair drivers' risk perception and decision-making abilities, thereby jeopardizing driving safety [15, 44].

Current research on emotional regulation in driving primarily employs feedback methods based on olfactory, visual, and auditory modalities. However, these approaches often face challenges such as inconsistent effectiveness or an increased cognitive load [20, 28, 41]. While tactile feedback has been widely applied in in-car interaction design and has proven its value in enhancing attention and decision-making in high-demand driving scenarios, its potential in emotional regulation remains underexplored [14, 19, 50]. We were inspired by the concept of "haptic creatures" in the field of Human-Robot Interaction (HRI), which are robots centered around haptic interaction and designed with pet-like characteristics [63]. These robots focus on the role of emotional touch in human-robot interaction and have demonstrated significant effectiveness in alleviating users' negative emotions [47, 62]. This approach, rooted in biological characteristics, provides a novel direction for emotional regulation in driving contexts.

Based on this research, we propose Miao-BOT, a pneumatic tactile steering wheel inspired by the design of a cat's paw. This design mimics the unique tactile patterns of a cat's claws and pads, aiming to create an "emotional connection" in the driving environment. The tactile feedback provided by Miao-BOT not only offers physical interaction but also fosters identity recognition and emotional transference by simulating the tactile characteristics of a cat. This enables drivers to experience emotional support in the relatively isolated driving environment, thereby alleviating emotional burdens. A comparison was conducted between the Miao-BOT-equipped steering wheel and a control group equipped with only basic tactile feedback, to evaluate the differing effects of their tactile feedback in regulating negative emotions and providing emotional companionship. Combining user feedback data and semi-structured interviews, we conducted an in-depth exploration of Miao-BOT's role in regulating negative emotions during driving.

The primary contributions of our study are as follows: (1) The design and implementation of the pneumatic haptic creature Miao-BOT on the steering wheel, which provides an innovative solution for regulating negative emotions in driving scenarios. (2) Through user studies and interviews, we analyzed and discussed the potential role of metaphor in shaping the tactile identity and emotional transfer of haptic organisms in driving scenarios, as well as the contextual event triggers required for effective haptic interaction.

2 RELATED WORK

2.1 Emotional Regulation in Driving: The Role of Tactile Feedback

Negative emotions experienced by drivers, such as fatigue, anxiety, and anger [29, 39, 48], can lead to hazardous behaviors, including distraction, delayed reactions, and poor decision-making [15, 44, 48, 51]. Research has shown that sensory feedback modalities such as olfactory [41, 47], visual [20, 22, 59], and auditory feedback [11, 20, 35] can help regulate emotions. However, their effectiveness is often limited by individual differences and may divert attention or increase cognitive load due to excessive information [35]. In contrast, in-vehicle haptic interactions, including those on seats and steering wheels [2, 19], offer greater privacy, faster response times, and lower levels of attention distraction [13, 35, 50]. Additionally, since human hands are more sensitive to tactile stimuli [38], haptic feedback on the steering wheel can better assist drivers in maintaining attention and enhancing decision-making under high cognitive load conditions [14, 34]. Despite these advantages, research on regulating drivers' emotions through tactile feedback remains limited. Therefore, building on the foundation of tactile interaction on the steering wheel, this study proposes a tactile feedback method designed to regulate emotions and alleviate drivers' negative emotional states.

2.2 Affective Haptics for Emotion Regulation

In the field of Human-Computer Interaction (HCI), various virtual and tangible technologies have been explored to sense and improve mental health [5, 27, 54, 60]. Among these, tangible interfaces offer significant opportunities for devices and systems aimed at enhancing emotional states, as they can both sense mental health conditions and provide physical intervention feedback [60]. Previous research has predominantly focused on emotion communication, while studies specifically targeting emotion regulation support remain underdeveloped [57]. Behavioral science has shown that affective touch can reduce stress and facilitate social interaction [17]. Affective haptics, such as tapping or stroking devices [42, 52], or receiving vibrations, warmth, and pneumatic pressure from them [8, 40, 56], serve as intuitive biofeedback methods that help individuals recognize intense emotional states and learn to self-soothe [12]. The effectiveness of these interactions is also closely tied to their social context. Price et al.'s study suggests that presence and identity are essential prerequisites for interpreting tactile communication [45]. In particular, the design of identity significantly influences whether users are willing to project emotions and thoughts onto robots, thereby attributing lifelike qualities to them [37].

These insights provide valuable references for designing affective tactile interaction systems to support emotion regulation. Our work aims to further explore the potential of leveraging identity to convey emotions, creating systems that enable meaningful emotional connections.

2.3 The Role of Tactile Robots in Emotional Regulation

We found that receiving information from robotic pets through visual and tactile modalities significantly influences human cognition

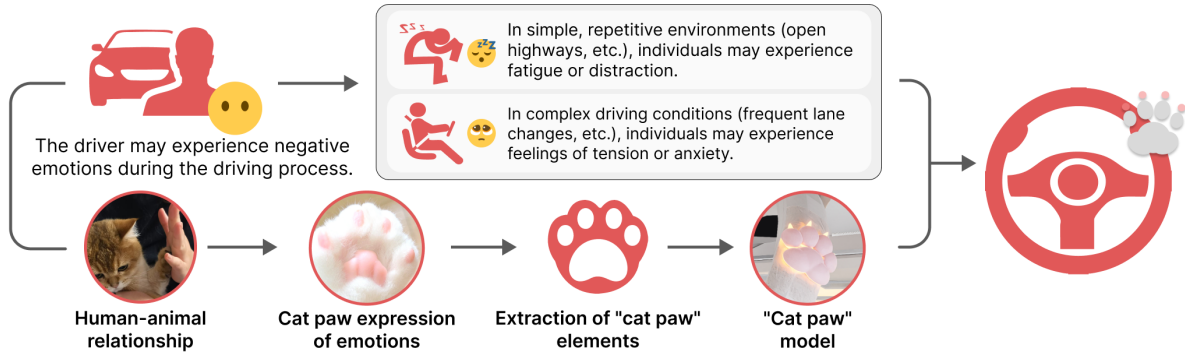


Figure 2: By extracting cat paw elements and incorporating human-animal relationships, emotional tactile feedback is provided to the driver to adapt to different driving scenarios.

and, in turn, alters behavior. Many robotic pets, such as the seal Paro [53], the robotic cat NeCoRo [32], and the robotic dog AIBO [18], as well as cuddling robots inspired by animals’ “snuggling behaviors,” such as SuggleBot [43], demonstrate that their soft tactile sensations create opportunities for fostering attachment and alleviating loneliness [3, 6, 26].

Furthermore, haptic creatures centered on touch—effectively highlight the role of emotional touch in social interactions [55]. A series of studies by Steve Yohanan and colleagues have shown that human emotional states are directly influenced by emotional touch interactions with robots [7, 61–64]. We observed that haptic creature robots with identifiable personas can effectively alleviate negative emotions. Thus, we aim to introduce them into driving scenarios to further explore their roles and effects in this specific context.

3 SYSTEM DESIGN

3.1 Design Motivation

Previous studies have highlighted that despite their diverse animal-inspired appearances and unique interactive features, haptic creature robots share a common characteristic: they enhance tactile semantics through biological imagery, enabling more natural and meaningful interactions between robots and users [1, 43, 65]. (Figure 2)

Emotional Regulation Needs in Driving Contexts: Our robot aims to alert drivers when they exhibit fatigue or distraction and provide comfort and emotional companionship during moments of tension or anxiety.

Selection of Biological Inspiration: Research has shown that, compared to other animals, cats exhibit a stronger emotional connection with humans [23, 46]. Specifically, the soft and delicate texture of cat paw pads provides a calming and relaxing effect, while the sharpness of their claws serves as a mechanism for warnings and alerts. Additionally, Yuhan Hu et al. demonstrated that circular “goosebumps”-like skin textures can convey positive emotions, whereas “spike”-shaped tactile surfaces are associated with negative emotional expressions [24, 25]. These findings align closely with the requirements of our tactile emotional design. Similarly, considering the impact of shape change frequency on arousal levels

[16], we designed distinct frequencies to simulate the movement characteristics and vitality of a cat’s paw. Low-frequency shape changes on rounded tactile surfaces were used to evoke a sense of calm, while high-frequency shape changes on pointed tactile surfaces were employed to enhance alertness.

Design Implementation: Based on these considerations, and combining the needs of the driving context with the characteristics of biological inspiration, we designed Miau-BOT—a haptic creature robot inspired by the cat’s paw. The tactile interaction is realized through pneumatic inflation and deflation modes:

- **Positive Comfort Feedback:** Circular “goosebumps”-like textures, representing the cat’s paw pads, provide a soothing tactile experience to alleviate drivers’ tension and anxiety.
- **Alert and Warning Functionality:** “Spike”-shaped textures, mimicking the sharpness of a cat’s claws, deliver distinct tactile alerts to warn and remind drivers.

The robot is mounted on the steering wheel to meet the diverse demands of negative emotion regulation in driving scenarios. Through tactile interaction, it enhances the safety and comfort of the driving experience.

3.2 Material and Manufacturing



Figure 3: Fabrication process of Miau-BOT’s interactive airbag system.

We utilized food-grade silicone with a hardness of 10 Shore A to fabricate Miau-BOT’s core interactive component: the airbag.

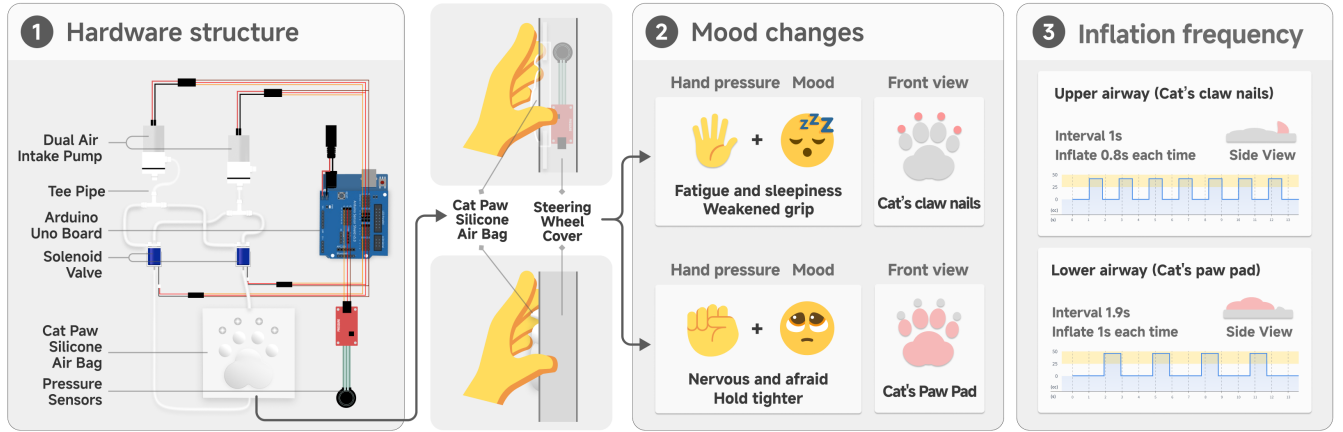


Figure 4: Design and development of Miau-BOT. (1) Core hardware structure diagram. (2) Tactile feedback patterns under different emotional states. (3) Design details of pneumatic tactile stimulation.

The airbags were produced using 3D-printed molds and feature two independent air channels. The upper channel is connected to claw-like spike airbags (“spikes”), while the lower channel is linked to pad-like airbags (“dots”). These two channels can be controlled independently, enabling different inflation patterns to enhance the tactile interaction experience. The prototype’s internal wiring is seamlessly integrated into a plush steering wheel cover, maintaining an organized and seamless appearance. The silicone airbags extend through openings in the steering wheel cover, serving as tactile elements for user interaction. (Figure 3)

3.3 Robot Hardware Development

In the hardware implementation, we used an Arduino Uno to control the pneumatic system, managing the operation of one inflation pump and one deflation pump. These pumps are connected to the air tubing via a three-way valve to ensure efficient distribution of air (Figure 4). Since pressure data from the steering wheel can largely reflect the driver’s emotional state and level of attention [4, 49], we incorporated pressure sensors as the hardware input during the user experiment phase, completing the initial hardware prototype design.

4 USER STUDY

We evaluated the impact of introducing the Miau-BOT steering wheel on drivers’ emotional regulation by recording participants’ performance and experiences while performing standardized pressure-inducing tasks in a simulated driving environment.

4.1 Participants

We recruited 16 participants (7 men and 9 women) from the college community, ages 19 to 40 (Mean=29.5, SD=10.5), with an average of 5.2 years of driving experience (SD=6.28), and eligible criteria required that they had no history of depression, anxiety, or cardiovascular disease (Table 1). Each participant was compensated with \$10. During the study, all participants provided informed consent.

Table 1: Demographic Information of Participants.

No.	Gender	Age	No.	Gender	Age
P1	M	40	P9	M	23
P2	F	24	P10	F	20
P3	M	21	P11	F	26
P4	M	32	P12	F	24
P5	M	21	P13	F	24
P6	M	23	P14	F	23
P7	M	20	P15	F	19
P8	F	38	P16	F	23

4.2 Procedure

We designed a controlled experiment to investigate: (1) the superior effectiveness of tactile bio-inspired feedback compared to dot matrix tactile feedback in regulating negative emotions; and (2) how Miau-BOT influences drivers’ emotions and driving behaviors under different driving scenarios.

Inspired by Marlene Weber et al.’s research on driving road scenarios and driver emotions [58], two distinct and typical driving scenarios were selected for the simulation videos: (1) Fatigue: Long periods of highway driving with minimal traffic, where evaluators reported feeling fatigued and distracted. (2) Tension: Driving in the emergency lane on the highway, involving continuous wrong-way driving, lane changes, and sudden braking. Evaluators experienced constant tension. We created a simulated driving environment for the participants in a soundproof meeting room free from auditory and visual distractions. For the experiment, we prepared two sets of steering wheels: (1) A dot matrix tactile feedback steering wheel (Figure 5 b-1), consisting of an exposed silicone airbag with geometric air channels. The silicone surface featured three dots and three spikes, providing the same functional feedback as Miau-BOT but without the cat paw design. (2) A Miau-BOT-equipped steering wheel, covered in a black plush material (Figure 5 b-2 and c), incorporating the tactile bio-inspired cat paw design.

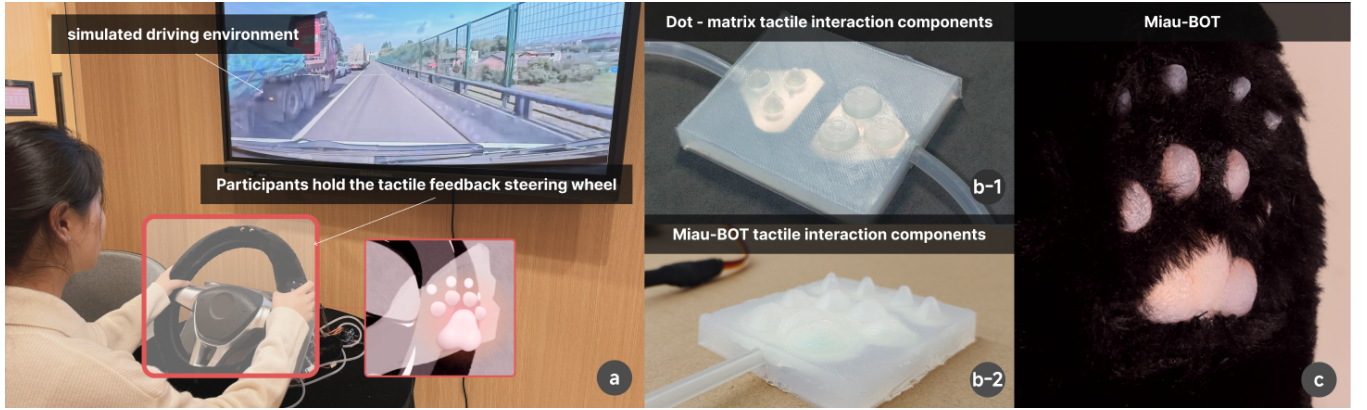


Figure 5: (a) User experiment setup. (b) Control experiment components with the same functionality, differing in that Miau-BOT features the cat paw image: (b-1) Dot matrix tactile interaction component; (b-2) Miau-BOT tactile interaction component. (c) The Miau-BOT component installed on the steering wheel.

The study was conducted as individual 50-minute sessions, co-hosted by two experimenters. Participants were asked to hold each steering wheel under identical driving scenarios, experiencing two predefined emotional contexts—fatigue and tension. After each session, participants completed a questionnaire, with all items rated on a 5-point Likert scale. Following the experiment, participants joined a 15-minute semi-structured interview to provide further insights.

5 RESULTS AND FINDINGS

5.1 Evaluation of Emotional Companionship

We conducted an independent samples t-test to evaluate the impact of two interaction forms on drivers' sense of companionship. Figure 6 shows that the Miau-BOT steering wheel received significantly higher scores compared to the dot matrix steering wheel (Miau-BOT steering wheel: 3.63, dot matrix steering wheel: 2.88; $p = 0.042 < 0.05$), indicating that the cat paw design provides a stronger sense of companionship. User feedback further supported this finding, suggesting that Miau-BOT not only offered better companionship but also provided emotional support that effectively helped regulate negative emotions. For instance, P5 and P12 remarked, "It feels like there's a cat keeping me company." Similarly, P11 noted, "Being able to receive reminders in a soft and intimate manner during driving is a very fitting emotional companionship experience."

We hypothesize that this emotional companionship experience stems from the emotional bond established between the participants and Miau-BOT.

5.2 Metaphors of Emotional Transfer and Tactile Identity

Through semi-structured interviews combined with open coding [21], we identified potential factors influencing the variability in emotional companionship experiences provided by Miau-BOT.

During the interviews, we observed that emotional connections with Miau-BOT were obviously reported among participants with prior pet ownership (P2, P3, P16). For instance, P2 remarked, "I used

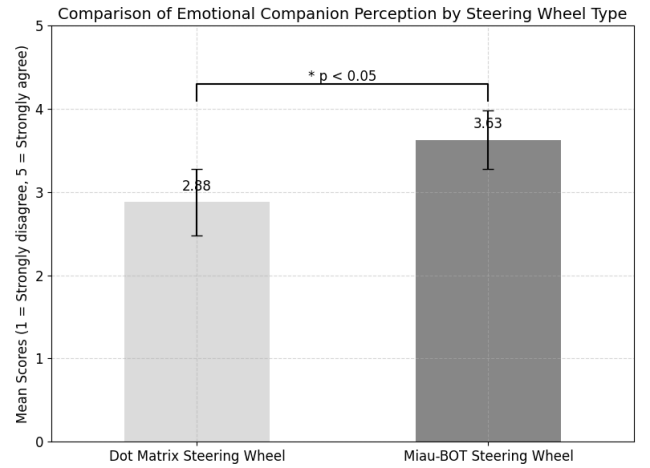


Figure 6: Analysis of the Companion Perception Questionnaire (Appendix A.1 Q4).

to have a puppy, so this animal-like design feels more relatable to me." This phenomenon aligns with attachment theory, which posits that individuals can establish emotional bonds with pets and project their emotional needs onto pets [31]. Research further suggests that emotional relationships between humans and pets can be transferred to robots, a process known as emotional transference [33]. Participants with pet ownership experience, having built strong emotional foundations with their pets, were more likely to transfer these emotions to Miau-BOT.

Similar phenomena of **emotional transfer** related to human-pet relationships were also observed among other participants. For example, compared to the dot matrix tactile device, the cat paw design made users more patient. P10 noted, "With the previous steering wheel device, I would feel irritated after prolonged use, but the cat paw design doesn't; it makes me feel more relaxed." Additionally, some participants found Miau-BOT to be more approachable. P13

commented, *“The tactile sensation of the cat paw feels better, and the design is really creative. It gives a sense of closeness, without the cold, robotic feel.”* Emotional transference helped bridge the gap between Miao-BOT and participants, fostering emotional connections.

Beyond the effects of emotional transference, tactile experiences further deepened emotional bonds. In this process, metaphor plays a crucial constructive role, allowing users to enhance their understanding through imaginative supplementation. Since individual experiences and cognition vary, the process of association and interpretation differs among users. The interpretation of the metaphor itself serves as a means for users to deepen their connection with the device. Studies indicate that **tactile identity**, which is shaped through tactile experiences, acts as a social cue for interpreting tactile meaning and can influence whether users are willing to project vivid emotions onto tactile objects [37], such as associating a warm and plush touch with *“a familiar pet”* [36].

We found that tactile identity played a positive role in strengthening human-machine emotional relationships when touch was the primary interaction mode. For example, P12 described Miao-BOT’s tactile feedback as *“feeling like a little cat exhaling softly.”* Similarly, P2, P7, and P11 mentioned sensations such as *“a breathing feeling”* and *“vibrations matching a heartbeat rhythm,”* which highlighted the warmth, rhythm, and vitality of the tactile interaction. P12 further commented, *“It gives me all the traits of a pet,”* indicating that the lifelike tactile feedback helped Miao-BOT establish a realistic and multidimensional pet-like identity. Additionally, P12 and P13 described Miao-BOT as *“a little bodyguard”* or *“someone who can help me steer,”* attributing it with a functional identity related to safety assurance. P2 and P6 referred to Miao-BOT as a *“friend”* or *“family member,”* emphasizing closeness, trust, and strong emotional bonds. The process of associating tactile experiences with tactile identity varied among participants due to differences in their personal experiences, but this process ultimately strengthened the emotional bond between humans and machines.

During driving, touch served as the sole and direct sensory channel for participants to perceive Miao-BOT, providing ample space for imagination and enabling participants to construct Miao-BOT’s identity. Research suggests that incorporating metaphors in design to leverage human empathy toward living beings can facilitate the establishment and maintenance of emotional connections [9, 10]. By assigning Miao-BOT a social identity, it entered the realm of emotional and social interaction, becoming a source of emotional support for users. In summary, emotional transference provided the foundation for Miao-BOT’s emotional connections, while tactile identity, reinforced through imaginative tactile experiences, strengthened these bonds, resulting in enhanced emotional regulation outcomes.

5.3 The Contextual Events of Tactile Emotion Regulation

Emotional transference and tactile identity provide explanations for Miao-BOT’s superior emotional regulation capabilities. On the other hand, our interviews revealed that the effectiveness of emotional regulation is also significantly influenced by the specific driving context. Research indicates that different scenarios impact

users’ interpretation of tactile feedback, resulting in varying emotional regulation outcomes [13].

In fatigue scenarios, the haptic feedback of the paw-inspired design did not provide effective emotional regulation. P2 stated, *“The tactile information from the cat paw doesn’t feel very noticeable when I’m fatigued, but it becomes more apparent in a tense state.”* Similarly, P5 suggested that Miao-BOT’s feedback might be more effective if it responded to specific events, further supporting the notion that when emotional needs are unclear, the effectiveness of haptic feedback diminishes. Conversely, in tense situations, the impact of haptic feedback was more pronounced. For example, P10 mentioned, *“I could feel myself relaxing in those tense situations,”* and P12 also noted, *“In specific scenarios like tension or distraction, I feel its feedback is more effective.”*

By comparing fatigue and tense scenarios, we found that users interpreted Miao-BOT’s haptic feedback based on their situational context. P12 used the term “randomness” to describe the fatigue scenario: *“Randomness gives me a sense of unease,”* indicating that the user could not identify a clear cause for the haptic feedback, suggesting the absence of a triggering event. Psychological fatigue accumulates gradually [30], and in the absence of a distinct triggering event, if fatigue has not reached a noticeable threshold, users may struggle to associate their fatigue state with the haptic feedback. This may ultimately lead to a loss of focus. For instance, P2 stated, *“The vibrations tend to distract me in such moments.”* P12 further emphasized, *“I would want this little cat’s sense of companionship to manifest when something actually happens,”* highlighting that users expect haptic feedback to be linked to specific events, making it a meaningful cue.

These findings suggest that the interpretation of tactile feedback is closely tied to contextual events. The effectiveness of tactile emotional regulation becomes more pronounced during scenarios involving clear emotional needs or distinct situational triggers. Driving scenarios in real-world contexts are far more complex and diverse [58]. Therefore, future research could leverage LLM-based contextual awareness technologies to further explore the contextual nature of tactile emotional regulation, enhancing its broader applicability and practical value.

6 LIMITATION AND FUTURE WORK

While this study innovatively proposed a tactile bio-inspired design for driving scenarios and validated its effectiveness, we must acknowledge certain limitations.

All participants in this study are from the same country and city, and the sample size is relatively small, with a greater emphasis on qualitative analysis. Future research will expand objective data analysis and include participants from diverse backgrounds. Although users rarely mentioned the impact of the additional fur and eye decorations on the Miao-BOT device during the experiment, we will strive to control variables more rigorously in future studies.

Currently, the detection of drivers’ emotional states relies primarily on pressure sensors, this single detection method is prone to inaccuracies. In the future, we plan to incorporate more diverse emotion detection methods, such as facial recognition and heart rate monitoring, to enhance the accuracy of identifying drivers’ emotional states.

7 CONCLUSION

This work investigates how haptic creatures can regulate drivers' negative emotions in driving scenarios. To address this, we developed Miau-BOT, a pneumatic robot equipped with tactile feedback featuring soothing dots and alerting spikes. We conducted a user study with 16 participants to evaluate emotional feedback to Miau-BOT compared to a dot matrix tactile device control group in both tension and fatigue scenarios. The results demonstrated that Miau-BOT, with its cat paw-inspired bio-identity, had a significant effect on emotional companionship. During emotional regulation, participants experienced stronger emotional support through metaphors of emotional transference and tactile identity. Moreover, the study revealed that the emotional regulation effect of tactile feedback was more pronounced when tension-inducing events occurred, triggering heightened emotional responses. This exploration provides valuable insights into the potential role of haptic creature robots in regulating negative emotions in driving contexts, offering a foundation for future designs in this area.

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A User Study Supplementary Information

A.1 Questionnaire

We employed a Likert scale¹ to quantitatively rate the following five questions, and participants completed the questionnaire after both the dot-matrix and the Miau-BOT experiments.

- **Q1:** Was the tactile sensation of the steering wheel clear? (1–5, “Very unclear” to “Very clear”)
- **Q2:** Was the intensity of the haptic feedback appropriate? (1–5, “Very inappropriate” to “Very appropriate”)
- **Q3:** Did the haptic feedback help you stay focused when distracted? (1–5, “Strongly disagree” to “Strongly agree”)
- **Q4:** Did the haptic feedback provide a sense of emotional companionship? (1–5, “Strongly disagree” to “Strongly agree”)
- **Q5:** Would you be willing to use this haptic steering wheel in the future? (1–5, “Very unwilling” to “Very willing”)

A.2 Semi-structured interviews

This study aims to investigate participants' perceptions of emotional companionship and driving experience while using different haptic steering wheels.

Haptic Feedback Perception

Could you describe the specific haptic sensations you experienced during the trial?

How did the haptic experience influence your preference for the steering wheel? In different scenarios (fatigue and stress), how did your perception of the haptic feedback differ between the two steering wheels?

Impact of the Paw-Inspired Steering Wheel on Driving

In different scenarios (fatigue and stress), how did the paw-inspired design affect your driving experience?

Did you feel that this design helped you maintain better driving performance?

Need for Emotional Companionship

Do you think Miau-BOT steering wheel could provide emotional companionship?

Pet Ownership Experience

Have you ever owned a pet? If so, do you feel that your experience with pet companionship influenced your emotional connection to the Miau-BOT steering wheel design?

¹Likert, R. (1932). A technique for the measurement of attitudes. *Archives of psychology*.